RF FINGERPRINTING FOR USRP DEVICE AUTHENTICATION

Using Deep Learning to identify and authenticate radio devices from their unique hardware signal — without passwords or software credentials

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Every radio transmitter in the world — whether it is a WiFi router, a military radio, a drone controller, or a software-defined radio like USRP — is manufactured with tiny, unavoidable hardware imperfections. These come from microscopic variations in components like the power amplifier, crystal oscillator, and mixer circuits during the manufacturing process.
These imperfections create a unique distortion pattern in every transmitted signal — like a fingerprint or DNA of that specific hardware. No two devices produce exactly the same signal, even if they are the same model from the same factory.

This is called Physical Layer Authentication — security at the radio hardware level, not the software level. It cannot be bypassed by stealing passwords because the fingerprint is in the physics of the device, not in any software.

Raw radio signal (IQ samples) received from a USRP device transmitting WiFi frames

Deep learning model analyses subtle hardware distortions in the signal pattern

Model identifies which of the 16 USRP devices sent this signal with high accuracy

The Traditional Way:
Today, devices prove their identity using:

- Passwords and PINs
- Digital certificates
- MAC addresses
- Cryptographic keys

The problem with all of these: They are all software. They live in memory. They can be copied, stolen, replayed, or spoofed. If an attacker gets hold of a device's certificate or MAC address, they can impersonate that device completely — and the network cannot tell the difference.

An attacker clones a device's entire software identity — firmware, keys, certificates. The network sees identical software. RF fingerprinting still rejects it because the hardware imperfections are different.

What is a USRP?
USRP stands for Universal Software Radio Peripheral. It is a software-defined radio — a research-grade hardware device that can transmit and receive any radio signal under software control.
Unlike consumer devices, a USRP gives full access to raw IQ samples — the unprocessed radio signal at the hardware level. This is why researchers use it for RF fingerprinting experiments.

ORACLE Dataset: genesys-lab.org/oracle
(16 USRP X310 devices were used as transmitters, Distances tested: 2ft, 8ft, 20ft, 62ft)

I — In-phase component The component of the signal aligned with a cosine reference wave. This is the real part.
Q — Quadrature component The component aligned with a sine wave, exactly 90 degrees offset from I. This is the imaginary part.
Together they form one complex number: sample = I + jQ

Imperfection	Cause	How it appears in IQ
DC Offset	Imperfect mixer chip	I and Q values have a non-zero mean
IQ Imbalance	Amplitude/phase mismatch	Constellation is rotated or distorted
Carrier Frequency Offset	Crystal oscillator not exact	Constellation slowly rotates over time
Phase Noise	Oscillator jitter	Constellation points appear smeared
Amplifier Nonlinearity	Power amplifier distortion	Harmonic distortion in signal

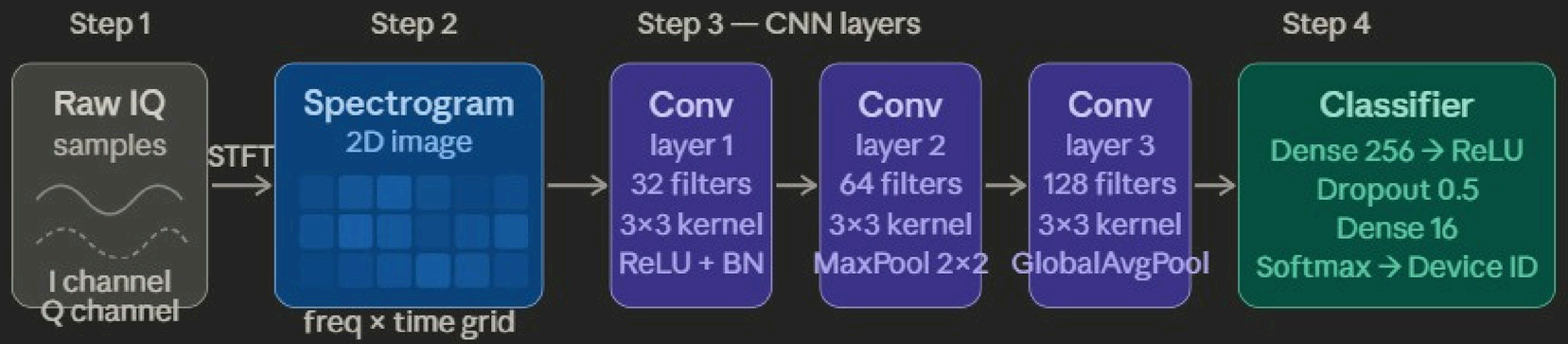
A CNN expects a 2D image as input. Raw IQ data is a 1D time series list of numbers. We need to convert it into a 2D image. We use STFT — Short-Time Fourier Transform
Step 1 — Take the raw IQ signal
Step 2 — Slide a window samples
Step 3 — Apply Fast Fourier Transform (FFT) inside each sample window. FFT converts time-domain data into frequency-domain data — it tells which frequencies are present in that window and how strong they are
Step 4 — Stack all the FFT results side by side to form a 2D image called a Spectrogram
What the spectrogram looks like:

- X-axis = time (each column is one window position)
- Y-axis = frequency (each row is one frequency bin)
- Pixel brightness = signal energy at that frequency and time

Hardware imperfections cause consistent subtle patterns in the spectrogram. Carrier frequency offset shifts where the peak appears. IQ imbalance creates asymmetric patterns. Phase noise blurs the peaks slightly. These subtle differences are consistent every time that device transmits
CNN input: Spectrogram image generated from STFT
LSTM input: Raw IQ time series directly, no spectrogram needed
Transformer Siamese input: IQ signal split into patches

Architecture 1 — CNN on Spectrograms

Here is exactly how the CNN approach works end-to-end:



What each layer learns from the spectrogram

Conv layer 1
Low-level signal edges
frequency onset patterns

Conv layer 2
Mid-level patterns
carrier offset, phase drift

Conv layer 3
Hardware fingerprint features
IQ imbalance, amplifier nonlinearity

Training setup

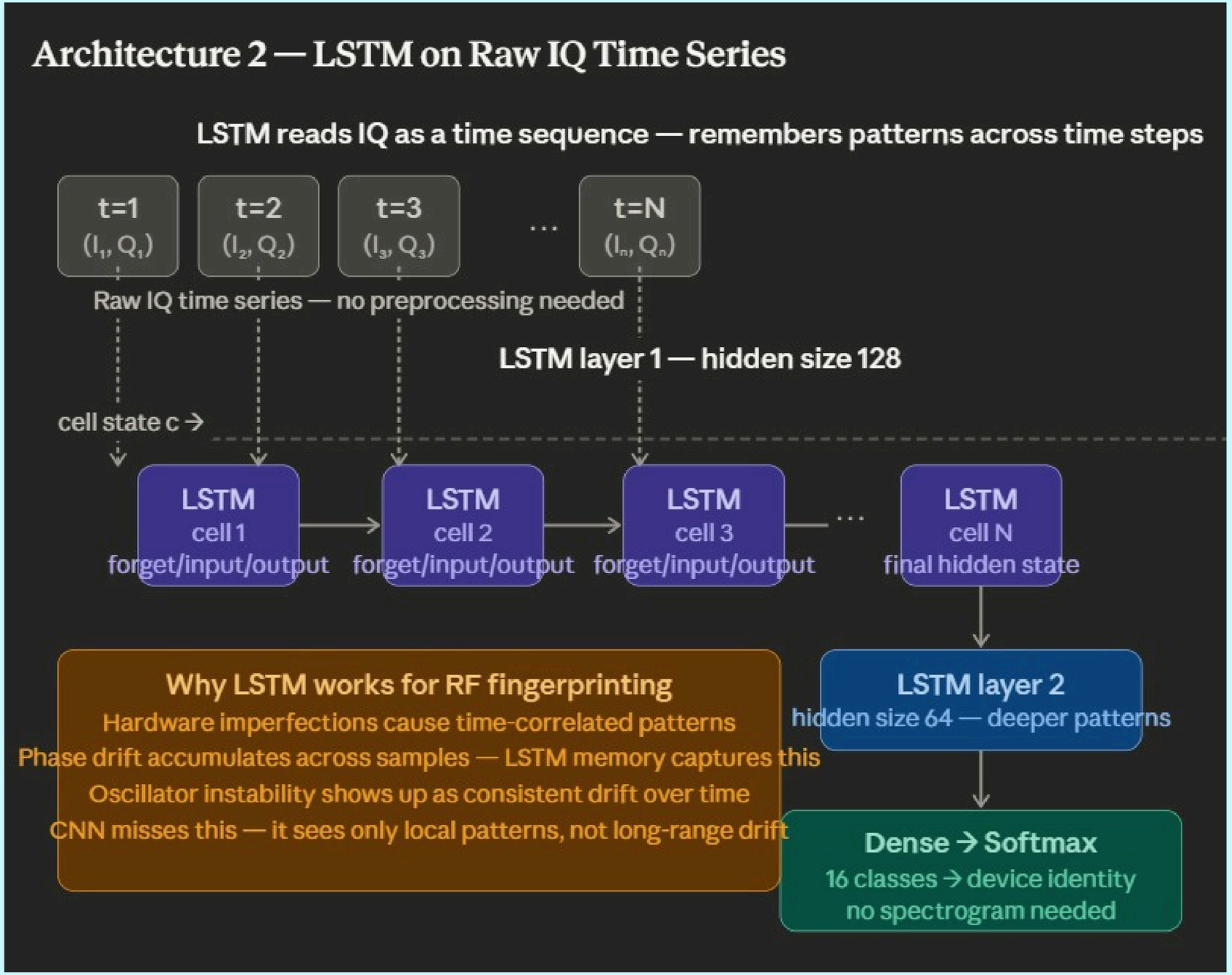
Data split
70% train
15% validation
15% test

Optimizer
Adam lr=0.001
Batch size 64
100 epochs

Loss function
Cross-entropy
16-class softmax
(one class per device)

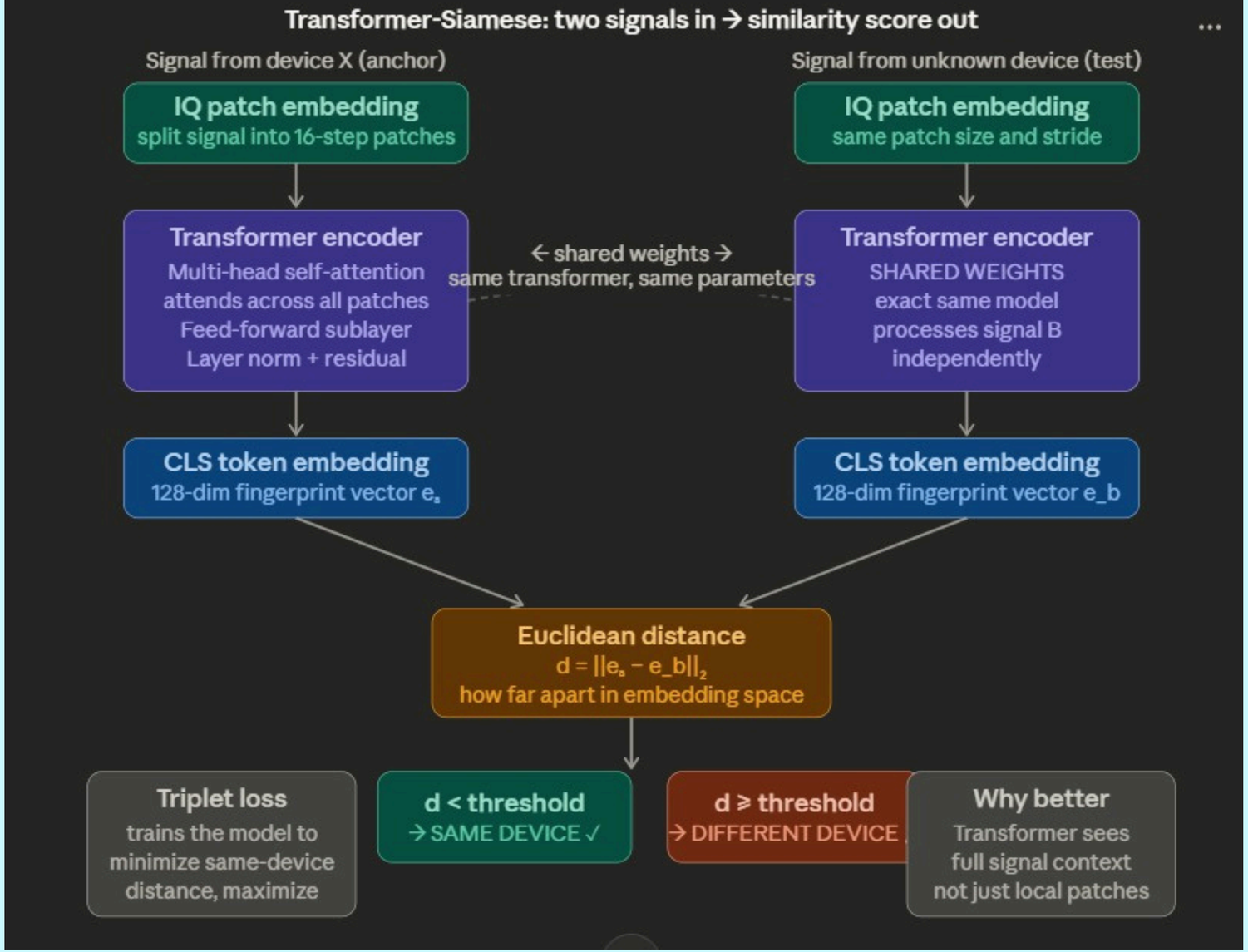
Expected result
~95% accuracy
on clean signals
your baseline

Architecture 2 — LSTM on Raw IQ Time Series



Architecture 3 — Transformer + Siamese Network (Your Novel Contribution)

This is the architecture that makes your paper publishable. Let me explain it in two parts — first the Transformer encoder, then the Siamese structure.



REAL LIFE APPLICATIONS

- Military and Defence Communications
- Unauthorized Radio Detection
- Drone and UAV Authentication

THANK YOU